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1. Defining “Research”

There are many definitions of research. A few are listed below.

Research is defined as the creation of new knowledge and/or the use of existing knowledge in a new and creative way so as to generate new concepts, methodologies and understandings. This could include synthesis and analysis of previous research to the extent that it leads to new and creative outcomes (University of Western Sydney, 2020).

Research is “a rigorous process of asking question, gathering data, analysing, interpreting, and proposing a response” to the question (Kinash, 2017).

“Any creative systematic activity undertaken in order to increase the stock of knowledge, including knowledge of man, culture and society, and the use of this knowledge to devise new applications” (OECD Glossary of Statistical Terms – UNESCO Definition, 2007).

Research allows for the advancement of a field of enquiry. This is known as research contributions and are made via the publication of thesis, dissertation, journal articles and various others. Research can be on a small scale for example a first year student trying to learn the Python programming language or on a much larger scale such as the Square Kilometre Array (SKA) project which brings together large resources and scientists from around the globe.

Without research great developments would not be possible like electricity, vaccines, nuclear fission and the internet. Without the ability to do simple research, students would not be able to decipher and complete simple assessments. In its simplest definition, research is a process of asking a question and then following a planned scientific approach to seek an answer to the question. Different researchers may arrive at different answers to the same research question because they have chosen a different scientific approach.

2. Basic types of research

The basic types of research are described below. The type is determined by the type of research question, the type of data to be collected and how it will be analysed.

Basic research: contributes to an existing body of knowledge through original results. It is *basic* in approach. It provides a deep insight into a problem and results in a scientific explanation and conclusion on it.

Applied research: utilizes research methodologies and techniques appropriate in the field of IT. It aims to solve a practical problem faced by society, industry or some other significant group. Experimental research, design research and action research as examples of applied research. This type of research could result in new techniques, products or methodologies.

Problem based research: aims to understand the *nature* of a problem and then find possible causes.

Qualitative research: aims at in-depth understanding of problems or issues in their *natural settings*. It is non-statistical. The data is collected from interviews, open-ended questions and interviews.

Quantitative research: uses computational and statistical techniques and calculations to collect and analyse data. Quantitative research is all about using numbers to prove or arrive at a conclusion.

3. The purpose and reasons for research

Research question and objectives determine the purpose of the research. Based on what the purpose is, research can conform to one of the following:

Exploratory research: is usually conducted as a first step to a larger research project. This type of research does not arrive at a conclusive result. Instead, it is used to establish an understanding of a research problem. After this, more focused research questions and objectives can be set.

Explanatory research: is used to better understand *relationships*. *Experiments* are a popular research strategy for this. For example, what has been the effect of using Moodle on first year IT students' performance?

Descriptive research: describes the characteristics of a population or phenomenon being studied. It does not answer questions about how, when and why the characteristics occurred. Instead it is directed toward answering questions such as, "What is this?"

Longitudinal research: involves data collection at several points over a long duration. For example, the determine the long term effects of smartphone misuse on the future workforce.

Cross sectional research: are mostly exploratory or descriptive in purpose. Data is collected and analysed once-off over a short period of fieldwork. It aims to look at *how things are now* and not pay attention to any trends.

The table below shows potential reasons for doing research and provides a brief explanation of this. This is adapted from Oates (2006).

<i>Reason for doing research</i>	<i>Brief explanation</i>
To add to the body of knowledge	Findings from a research study within a field of IT. This new knowledge may not solve a persistent problem but, it will <i>contribute</i> towards finding one.
To solve a problem	Develop a new supervised machine learning algorithm specific to a problem e.g. lung cancer detection.
To find evidence to inform practice	How do we? How should we? To discover better methods, models and algorithms. Then evaluate current with new to see which is better. To test the circumstances under which it is better. Good research must provide evidence of whether or not the new is better.

To develop a better understanding of people as they interact with the world	This in turn helps develop IT solutions that are better suited or fit-for purpose. E.g., improving the UX on mobile devices, removing less used components from a website.
To test or disprove a theory	There are many existing theories. New theories can be formulated. Through a research process these can be approved or disapproved.
To develop a better way	Improve an existing, well established way to solving a problem e.g. Object orientated programming, agile methodologies

4. Research in the field of IT

Research has led to great advancements in IT. Some of the greatest developments include:

- Alan Turing. By 1932, he was writing mathematical whitepapers concerning algorithmic truths and conceptualizing Turing Machines as the earliest computer.
- The concept for the 'mouse' was invented in 1941. Secretly patented in 1946 by the British Royal Navy as the 'ball tracker'.
- Intel developed the first micro-sized central processing unit (CPU), the 4004, in 1971. It could process four bits at any given time.
- The first portable computer 1973, the Osborne 1 boasted a 4MHz CPU, 64 kilobytes of memory, a detachable (but wired) keyboard that also attached itself as part of the protective casing for transport and a built-in 5-inch black-and-white screen.
- In 2002, Intel revolutionized another aspect of computer processing through hyper-threading. In each processor there are a number of cores and hyperthreading parallelizes computing of multiple tasks.
- The main identifier of artificial intelligence is that the machine can think freely and critically rather than just delivering programmed responses to standardized inputs. Turing developed three ways to test for actual artificial intelligence in a computer 50 years before we saw a machine that possessed even the most rudimentary AI capabilities. Nothing has managed to pass Turing's test yet.

(Isaacs, 2016)

Tutorials

1. Section 3 describes the purpose of research. For each of the following, state which purpose is most likely:
 - a. To determine the impact of excessive online meetings on the office worker's psychology
 - b. Measuring the impact of online learning tools on student's performance for an assessment
 - c. What are the challenges of mobile app use among the elderly?
2. List and briefly describe one recent great development in IT.
3. Can one research study be both quantitative and qualitative? Motivate your response by means of an example.

Useful urls

<https://www.scribbr.com/methodology/types-of-research/>

<https://www.questionpro.com/blog/what-is-research/>

References

- Isaacs, D. 2016. 10 biggest moments in computing history. Techradar. Available at:
<https://www.techradar.com/news/computing/10-biggest-moments-in-computing-history-1326851>
- Oates, B. J. 2006. Researching information systems and computing. Sage, London.